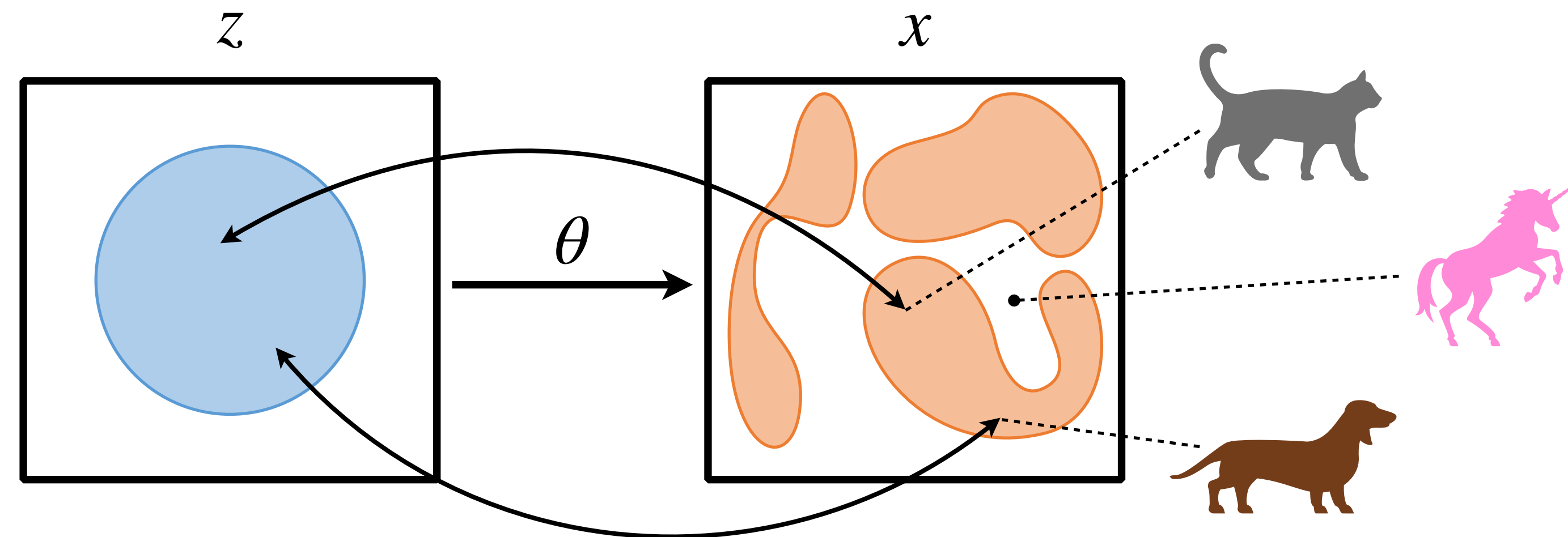


Invertible Neural Networks

Likelihood: $\mathcal{L}(\theta \mid x) = p_{\theta}(x) = P_{\theta}(X = x)$

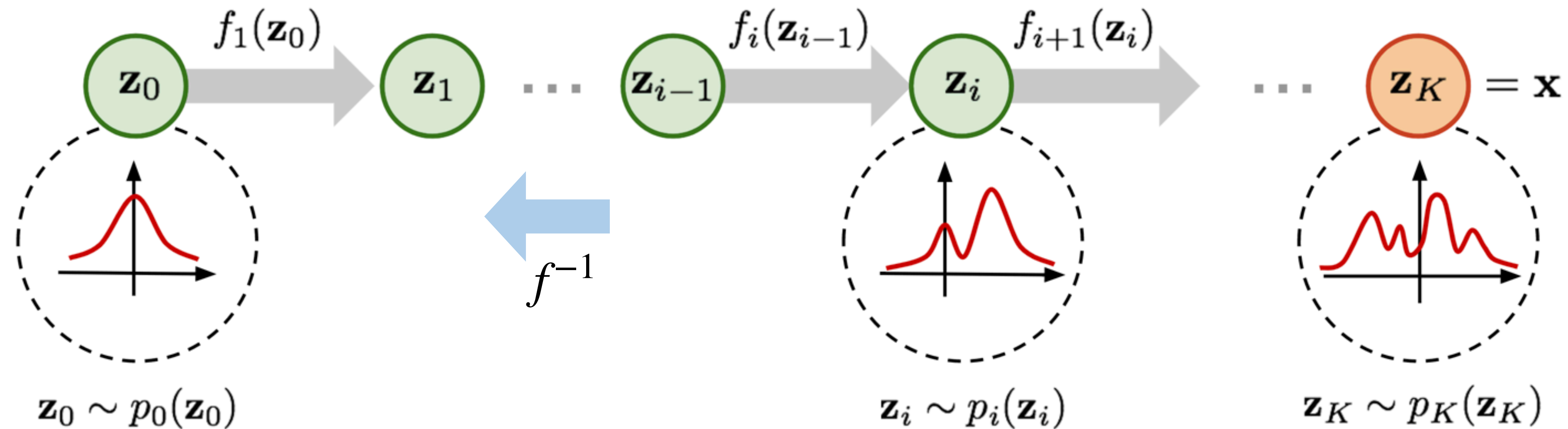
Probability of observing data x , assuming θ is the actual parameter.

$$p_{\theta}(x) = \int p_{\theta}(x, z) dz \quad \text{Intractable due to "combinatorial explosion" of } z$$



Invertible Neural Networks

<https://lilianweng.github.io/posts/2018-10-13-flow-models/>



$$\mathbf{z} \sim \pi(\mathbf{z}), \mathbf{x} = f(\mathbf{z}), \mathbf{z} = f^{-1}(\mathbf{x})$$

$$p(\mathbf{x}) = \pi(\mathbf{z}) \left| \det \frac{d\mathbf{z}}{d\mathbf{x}} \right| = \pi(f^{-1}(\mathbf{x})) \left| \det \frac{df^{-1}}{d\mathbf{x}} \right|$$

Gaussian
Inverse
Jacobian Determinant